

MODEL QUESTION PAPER

FOURTH SEMESTER DIPLOMA EXAMINATION IN INSTRUMENTATION ENGINEERING

MICRO CONTROLLER AND INTERFACING

TIME:3Hrs
Maximum Marks:100

PART -A

(Maximum Marks:10)

(Answer the following in one or two sentences.Each question carries 2 marks each)

I.

1. List the 16 bit registers in 8051
2. Mention the use of EA signal in 8051
3. Which bits of PSW are responsible for register bank selection
4. State the function of inservice register in 8259
5. Define conversion time (5x2=10)

PART -B

(Maximum Marks:30)

(Answer any five questions from the following .Each question carries 6 marks)

II.

1. Explain the function of IE and IP register
2. Draw the structure of PCON register and explain
3. Describe the serial data reception
4. Write an assembly language program to divide two hexadecimal numbers
5. Draw the pin diagram of 8279
6. Explain the registers of 8259
7. Describe the interfacing of D/A converter (5x6=30)

PART -C

(Maximum Marks:60)

(Answer one full question from each module. Each question carries 15 marks)

MODULE I

III.a). Draw and explain the pin diagram of 8051 15

OR

IV a) Draw and explain the structure of SCON register 8

b) Describe the features of 8085 7

MODULE II

V a) Classify the instruction set of 8051 and Explain with example 15

OR

VI . a) Write an assembly language program to add two 16 bit hexadecimal numbers 7

b) Write an assembly language program to sort an array in ascending order 8

MODULE III

VII. a) Illustrate the ports of 8255 with neat diagram 7

b) Describe architecture of 8279 8

OR

VIII . a) Describe the key board debounce and control 7

b) Explain block diagram of 8251 8

MODULE IV

IX. a) Describe the interfacing of lcd display 8

b) Describe the interfacing of A/D converter 7

OR

X. a) Describe the interfacing of LM 35 8

b) Describe the interfacing of stepper motor 7